

Using Machine Learning to Track the Onset of Alzheimer’s Disease

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Abstract

Alzheimer’s disease is a progressive neurodegenerative condition often diagnosed in its later stages due to the invasive and costly nature of standard clinical screenings. This paper proposes a novel, scalable forecasting framework that shifts from traditional population-level classification to individual-centric monitoring of cognitive decline. The project would utilise Long Short-Term Memory neural networks to predict future performance on the Clock Drawing Test by leveraging high-frequency, passive data from wearable devices, including gait patterns, sleep metrics, and circadian rhythms. To ensure objective metrics, a Convolutional Neural Network is employed to standardise and score clock drawings into interpretable sub-scores. By applying individual-level normalisation against a participant’s own historical baseline, the model will identify subtle behavioural deviations that precede measurable impairment. This integration of longitudinal behavioural data and automated cognitive assessment provides a non-invasive, highly-scalable, probabilistic tool for early risk assessment and personalised tracking of disease progression.

1 Introduction

Alzheimer’s disease (AD) is a progressive neurodegenerative disorder and the leading cause of dementia worldwide. It is characterised by the accumulation of amyloid- β plaques and tau neurofibrillary tangles, resulting in synaptic dysfunction, neuronal loss, and widespread brain atrophy. Clinically, AD manifests as a gradual decline in cognitive function, characterised by slight impairments in memory and executive functioning before progressing to more severe symptoms that interfere with daily life. A defining feature of Alzheimer’s disease is its long preclinical phase, during which changes in brain chemistry accumulate for years or even decades before clear cognitive symptoms are observed. This benign period presents a major challenge in early detection and prevention.

Current diagnostic and monitoring approaches for AD rely heavily on biological biomarkers and structured cognitive assessments. Neuroimaging techniques such as magnetic resonance imaging (MRI) and positron emission tomography (PET) provide detailed insights into structural brain changes and amyloid or tau deposits, while cerebrospinal fluid (CSF) biomarkers provide pathological information^{1–3}. Although these methods have significantly advanced Alzheimer’s research, they are expensive, invasive, and typically collected at infrequent time points. This limits the suitability for continuous monitoring, large-scale screening, or modelling individual disease trajectories over time.

1.1 Cognitive Assessments

Cognitive assessments remain a quick and non-invasive method to identify adults at risk of AD, with lower test scores strongly correlated with disease presence and progression⁴. Tests such as the Mini-Mental State Examination (MMSE) and Montreal Cognitive Assessment (MoCA) are widely used to assess global cognitive function⁵. However, these methods are criticised for their limited sensitivity to early-stage decline, educational bias, and practice effects⁴. Importantly, while single cognitive scores may be noisy and prone to bias, longitudinal patterns of performance tend to reflect meaningful cognitive change, making cognitive tests particularly valuable when analysed as trajectories rather than isolated events⁶.

Among commonly used cognitive assessments, the Clock Drawing Test (CDT)⁷ has been extensively validated as a sensitive marker of early cognitive dysfunction^{8,9}. It evaluates cognitive skills often affected early in Alzheimer’s disease, including visuospatial ability, executive function, planning, and cognitive control⁷ (examples shown in Fig.1). Compared to more technical tools, the CDT is quick to administer, less influenced by language or education, and has low false-negative/positive rates¹⁰.

Metrics have been developed to standardise CDT assessment: structural metrics describing clock integrity and number placement¹¹, As well as error-based metrics, recording omissions, misordering, or conceptual errors¹². These provide interpretable discrete targets for scoring and predictive modelling. However, the suitability in the CDT to differentiate between different types of dementia has been shown to be limited¹³.

Recent advances in tablet-based test administration

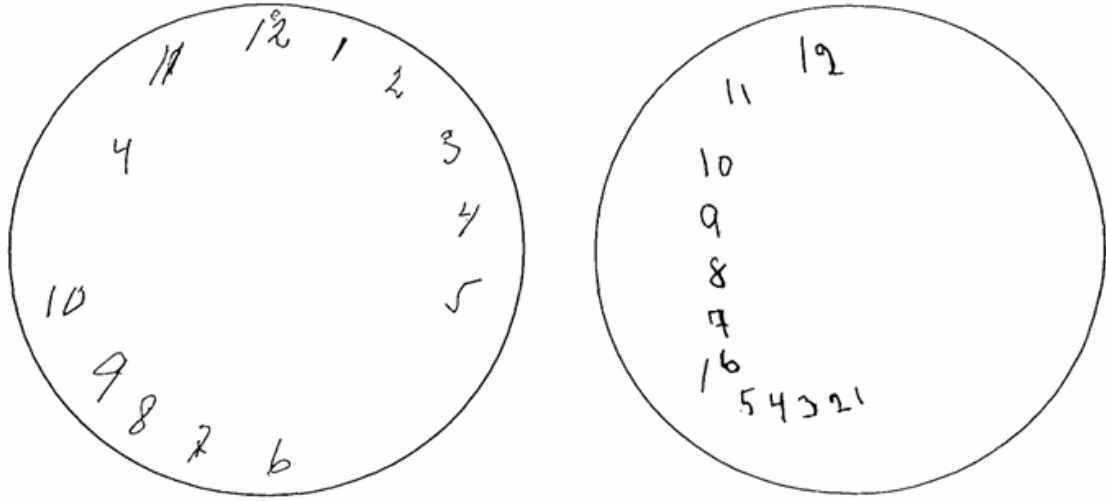


Figure 1: Two such examples of the Clock Drawing Test, both exhibiting a number of errors with incorrect numeral allocation and placement, demonstrating a lack of planning and mental coordination⁷.

enable more detailed collection including: stroke order, pauses, corrections, and pressure patterns¹⁴. This data provide a richer depth that can be analysed in isolation to reproduce the final test scores¹⁵.

1.2 Wearable Monitoring

While cognitive assessments capture the functional consequences of neurodegeneration, they do not provide insight into the behavioural changes that precede measurable cognitive decline. Growing evidence suggests that Alzheimer’s disease is accompanied by early alterations in physical behaviour and physiology, including gait slowing and increased stride variability¹⁶, physical activity levels. Similarly, sleep disruption and circadian instability are increasingly recognised as early features of the neurodegenerative process and have been linked to both amyloid accumulation and cognitive decline^{17;18}.

Behavioural changes often emerge gradually and vary between individuals, making them difficult to detect through irregular clinical observation. Advances in wearable sensor technology now allow continuous, passive, and scalable measurement of traits such as activity, heart rate, and sleep¹⁹. Unlike traditional assessments, wearable data enable long-term monitoring at the individual level. This approach, known as digital phenotyping, has gained traction in neurodegenerative disease research^{20;21}, with activity and sleep patterns differing between cognitively healthy individuals and those with Alzheimer’s or other cognitive impairments²². While laboratory-grade devices offer higher precision, wearables provide a practical trade-off between accuracy and scalability for longitudinal monitoring.

1.3 Machine Learning Approaches

Machine learning (ML) provides a flexible framework for modelling complex, high-dimensional data and has

been widely applied in Alzheimer’s disease and dementia research^{2;23}. Many studies utilise large longitudinal datasets, such as the Alzheimers Disease Neuroimaging Initiative (ADNI) or electronic health records, combining imaging, cognitive assessments, and clinical data to identify disease biomarkers and estimate risk^{23;24}. Traditional supervised models, including Random Forests, Support Vector Machines and Gradient Boosting, perform well at classifying disease status or short-term risk across groups, but often cannot provide reliable predictions for individual patients^{24;25}. Additionally Convolutional Networks are widely used to analyse brain scan data and detect bio-markers of disease far earlier than doctors could²⁶.

More recent work has shifted toward modelling disease progression using longitudinal data, employing sequence-based architectures such as recurrent neural networks (RNNs) and long short-term memory (LSTM) networks^{27;28}. These models are designed to capture non-linear temporal dependencies and delayed effects, making them well suited to understanding how cognitive performance and other disease markers evolve over time. At the same time, many of these studies rely on sparsely sampled clinical data or focus on population-level trends rather than the patterns of change experienced by individual patients.

Taken together, these developments highlight the potential of sequence-based modelling to capture evolving patterns over time, particularly when rich, longitudinal behavioural data is available.

1.4 Individual Progression

Alzheimer’s disease progression is highly heterogeneous, with substantial variability in onset, symptom manifestation, and rate of decline²⁹. Models trained to predict average outcomes across populations may therefore obscure early, individual-specific deviations that are clinically meaningful. In contrast, individual-centric forecasting approaches aim to model within-

person change relative to a personal baseline, aligning well with the time-rich data provided by wearables.

Framing Alzheimer’s disease modelling as a forecasting problem rather than a classification task represents a shift in how progression is studied. Rather than asking whether an individual currently meets diagnostic criteria- as would qualify for a scan- forecasting approaches seek to predict how cognitive function will evolve over time. Within this framework, cognitive assessments such as the CDT can be treated as a stable, interpretable outcome summarising functional decline and acting as a proxy for the biological truth, while wearable-derived behavioural features act as dynamic predictors reflecting underlying neurodegenerative processes.

2 Methodological Overview

Building on this individual-centric forecasting, the present study investigates whether longitudinal wearable-derived measures of activity, gait and sleep can be used to predict future cognitive assessment outcomes. By focusing on non-invasive, scalable data sources and temporal machine learning models, this work aims to complement existing biomarker-based approaches rather than replace them. Specifically, it seeks to address the gap between high-resolution behavioural monitoring and clinically meaningful cognitive outcomes.

This study proposes a longitudinal framework for forecasting cognitive decline associated with Alzheimer’s disease using passive wearable sensor data and repeated cognitive testing. The core hypothesis is that high-frequency behavioural data captured via wearable devices can predict future performance on sensitive cognitive assessments (specifically the CDT) before clinically significant decline becomes apparent. By using LSTM networks individual trajectories can be synthesised rather than relying on population averages.

2.1 Participants and Data Collection

Participants will be older adults at risk of cognitive decline(60+). Inclusion criteria will prioritise individuals capable of independently wearing a smartwatch and completing brief cognitive assessments with broadly unaffected vision and motor systems. No prior diagnosis of Alzheimer’s disease will be required at baseline, allowing the model to capture preclinical and early-stage changes.

Each participant will be monitored continuously using a standardised smartwatch capable of collecting accelerometry, heart rate, and sleep-related data. Wearables are selected due to their scalability and non-invasive nature. Cognitive testing will be conducted at regular intervals (e.g., monthly), so that wearable-derived behavioural patterns can be learned. The testing must be relatively infrequent to ensure experimental validity(participants over-learning the test) and to

minimise participant time taken. This will allow the study to expand greatly in cohort size.

2.2 Data Preprocessing and Feature Construction

Raw wearable data is inherently noisy and heterogeneous across individuals, necessitating careful preprocessing prior to modelling. Initial cleaning includes the removal of implausible physiological values and the identifying when the devices are not worn.

Given that cognitive outcomes (CDT scores) are assessed monthly, high-frequency wearable data will be collated into summary features capturing activity, gait, sleep, and circadian behaviour. This reduces the influence of sensor noise and mitigates gaps due to no data.

To maintain data quality, participants with insufficient coverage will be excluded based on minimum thresholds for valid days per month and overall monitoring duration. This filtering is justified by the scalable nature of wearable-based monitoring and the testing framework.

From the cleaned and aggregated data, features are extracted over fixed temporal windows and normalised relative to each participant’s personal baseline. This individual-level normalisation emphasises within-person change rather than between-subject differences, allowing subtle changes in behaviour to be detected despite large inter-individual variability in lifestyle, fitness, or baseline activity levels. This step is critical for isolating individual trajectories of cognitive decline and supporting personalised forecasting³⁰.

2.3 Cognitive Ground Truth: CDT

The CDT is used as the primary cognitive outcome, with assessments administered either via paper or digitally on a tablet. To ease scoring inconsistencies and increase experimental scale, CDT outputs will be processed using a convolutional neural network (CNN) trained to generate standardised scores from digitised drawings. This task is highly suitable to a CNN as the positional data is key to the scores and hand written detection is one of the most famous CNN based problems³¹.

Rather than relying on a single coarse summary score, the CNN will produce a small set of interpretable sub-scores reflecting distinct aspects of task performance, including:

- Overall structural integrity of the clock face.
- Spatial organisation and placement of numerals.
- Correctness of numerals
- Accuracy of hand placement and time representation.

These sub-scores could be combined into composite metrics or modelled separately, allowing patterns

of cognitive decline to be captured across the different skills assessed. This enables more sensitive comparison between outputs and specific behavioural features modelled by the LSTM and may provide additional sensitivity to subtype-specific or individualised progression patterns.

Where tablet-based assessments are utilised, the previously discussed process-level measures (drawing latency, stroke order, and pausing behaviour) will be extracted and incorporated as additional outcome metrics. These features will be treated as optional enhancements, ensuring the approach remains compatible with traditional paper-based testing and does not exclude participants on the basis of technological access or familiarity.

2.4 Temporal Modelling with LSTM Networks

To model the relationship between longitudinal behavioural data and future cognitive performance, a LSTM-based sequence modelling framework will be used. LSTMs are well suited to this task due to their ability to capture long-term temporal patterns, non-linear behaviour, and delayed effects, all of which are key characteristics of neurodegenerative disease progression.

For each participant, sequences of wearable-derived features over preceding months/years will be used as model inputs, with the prediction target defined as the CDT performance at a future assessment point (e.g., one or two months ahead).

The model will be trained in a supervised manner using rolling temporal windows, generating multiple training samples per individual. Rather than learning a single population-average trajectory, the model will learn predictions on each individual’s historical data, allowing deviations from personal norms to be associated with the following individuals cognitive trajectories.

2.5 Model Evaluation and Validation

Model performance will be assessed using standard regression metrics, including mean absolute error (MAE) and root mean squared error (RMSE), which quantify how accurately future CDT scores are predicted. Evaluation will use subject-to-subject cross-validation, ensuring that the performance will also extend to an unseen individuals activities rather than learning person-specific patterns.

3 Expected Results

As CDT cannot effectively differentiate between subtypes¹³, this study does not anticipate it serving as a sensitive diagnostic marker for the biological ground truth. Rather, the expected outcome is that longitudinal CDT trajectories, when modelled alongside

wearable-derived behavioural data, will further compound the accuracy of successive cognitive tests on detecting neurological deterioration⁶ and offer a more informative and sensitive representation of an individuals disease progression. Importantly, the aim is not to produce a single predictive score, but to generate individualised, probabilistic trajectories that capture the heterogeneity of neurodegenerative disease progression with personalised, longitudinal monitoring.

Neurodegenerative diseases often manifest subtle behavioural and motor changes prior to detectable cognitive decline. Previous studies have reported associations between early alterations in physical activity, gait dynamics¹⁶, sleep patterns¹⁸ and subsequent cognitive deterioration. Building on this, the study hypothesises that features related to gait variability, motor slowing, and sleep fragmentation will serve as early indicators of future CDT decline. The relative timing and importance of these features are expected to differ across individuals, reflecting the heterogeneous nature of disease progression (Figure 2).

Consistent with prior work demonstrating the advantages of temporal modelling in neurodegenerative disease research, the LSTM-based forecasting framework is expected to outperform static machine learning approaches also using wearable devices^{25;27}. By integrating historical behavioural and cognitive data, the model can capture delayed and cumulative effects that precede measurable cognitive decline³², generating personalised probabilistic forecasts for each participant.

However, there is no real likelihood that this method will serve to be more effective than regular scans at picking up early signs of Alzheimer’s and as such the model will have to run for a long period(at least two years) in order to accurately reconstruct behavioural norms and start to see deterioration within the cohort.

3.1 Clinical Interpretation and Feedback

Although the study is not designed as a diagnostic tool, the predicted trajectories could be readily translated into clinically meaningful insights for individuals. Rather than presenting raw CDT predictions, model outputs are expected to be most useful when framed as probabilistic trends: such as maintaining stable function, a gradual decline, or an accelerated deterioration- relative to an individual’s own historical baseline. This approach aligns with clinical reasoning and avoids overconfidence in single-point estimates.

A single CDT score in isolation provides limited information, and is prone to perturbing effects³³. In contrast, trajectory-based forecasts offer early warning signals by identifying subtle deviations from personal baselines before clinically significant changes occur. When an elevated likelihood of deterioration is inferred, this may support earlier monitoring or referral for neuroimaging and specialist assessment.

The integration of wearable data also enables identi-

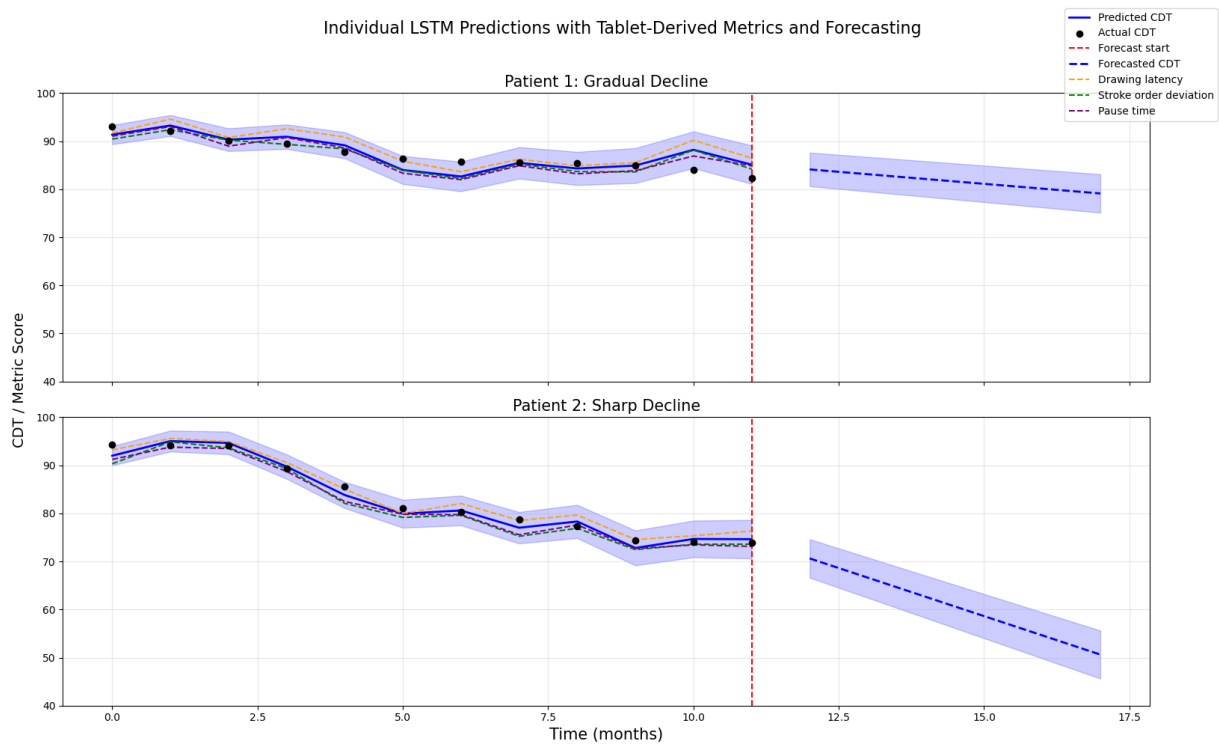


Figure 2: Illustrative example of expected model outputs, showing two distinct CDT trajectories with differing rates of decline and associated behavioural features derived from wearable data. Points represent CDT results, with the blue line acting as the prediction. Shaded regions represent predictive uncertainty with other coloured lines representing metrics gleaned from smart tablet data. After the 1 year (11 months) mark a predictive forecast for CDT results is plotted.

fication of abnormal progression patterns, such as cases where motor decline is evident without cognitive deterioration, or vice versa. These insights are intended to augment, rather than replace, clinical judgement by vastly extending the temporal and behavioural information available to clinicians when assessing patients.

Taken together, the expected clinical outcome of this work is improved individual-level risk stratification and earlier, more personalised interpretation of disease progression under uncertainty.

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